

Project Context: Multilingual Grapheme-to-Phoneme (G2P) with Phoneme Clustering

What This Project Does

This project builds a **multilingual Grapheme-to-Phoneme (G2P) system** for Indian languages (Hindi, Gujarati, Marathi), then improves it by **clustering similar phonemes** to reduce the output vocabulary. G2P models convert written text into phoneme sequences — a core component of Text-to-Speech (TTS) and Automatic Speech Recognition (ASR) systems.

Problem Being Solved

1. Most G2P systems train a **separate model per language** — this project explores one unified multilingual model.
2. Many phonemes across languages represent the **same sound** but are treated as distinct units, inflating the vocabulary unnecessarily.

Example: Hindi (क), Gujarati (ક), and Marathi (क) all produce (/k/) but are typically treated as separate phonemes.

Languages

- Hindi (hi)
 - Gujarati (gu)
 - Marathi (mr)
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Dataset

IndicTTS Dataset — <https://www.iitm.ac.in/donlab/indictts/database>

Contains audio recordings + text transcripts for multiple Indian languages. Processed into a pronunciation lexicon:

word → phoneme sequence

Final Dataset Format

word	phonemes	language
भारत	b aa r ax t	hi
ભારત	b aa r ax t	gu
भारत	b aa r ax t	mr

Pipeline (8 Phases)

Phase 1 — Data Preparation

Download IndicTTS → extract transcripts → normalize/clean → tokenize → phonemize → build pronunciation dictionary → merge multilingual dictionaries with language tags → train/val/test split.

Phase 2 — Baseline Multilingual G2P Model

Train a sequence-to-sequence model (Transformer or BiLSTM with attention).

- **Input:** `<language_tag> + word` (e.g., `<hi> भारत`)
- **Output:** phoneme sequence (e.g., `b aa r ax t`)
- **Metrics:** Phoneme Error Rate (PER), Word Error Rate (WER)

Phase 3 — Phoneme Inventory Extraction

Extract all unique phonemes across all languages.

Example inventory: `{p, b, t, d, k, g, kh, gh, aa, ax, e, i, o}`

Phase 4 — Phoneme Embeddings

Represent each phoneme as a numeric vector to capture similarity.

Example: `k → [0.32, 0.41, -0.18]`, `g → [0.35, 0.38, -0.17]`

Phase 5 — Phoneme Clustering

Cluster similar phonemes using K-Means or Hierarchical Clustering.

Cluster	Phonemes
C1	k, g, kh
C2	p, b
C3	t, d

Phase 6 — Replace Phonemes with Cluster IDs

Convert phoneme outputs to cluster labels to reduce vocabulary size.

- Original: b aa r ax t
- Clustered: C2 C8 C5 C7 C4

Phase 7 — Retrain G2P Model

Retrain the model using cluster IDs as output tokens. Expected benefits: smaller output space, faster training, better cross-language generalization.

Phase 8 — Evaluation

Compare baseline vs. clustered model.

Model	PER	WER	Output Vocab Size
Baseline	~12%	~18%	~80 phonemes
Clustered	~10%	~16%	~40 clusters

Additional metrics: model size, training time, cluster visualizations.

Tech Stack

Category	Tools
Language	Python
ML Frameworks	PyTorch, TensorFlow
Data & Math	NumPy, Pandas, Scikit-learn

Category	Tools
Visualization	Matplotlib, UMAP, t-SNE
Speech/Phoneme Tools	Phonemizer, Epitran, Montreal Forced Aligner

Key Terms

Term	Meaning
Grapheme	Written character
Phoneme	Smallest unit of sound
G2P	Grapheme-to-Phoneme conversion
Phoneme Inventory	Complete set of phonemes in a dataset
Embedding	Vector (numeric) representation of a symbol
Clustering	Grouping similar items together
PER	Phoneme Error Rate
WER	Word Error Rate

Expected Deliverables

1. A multilingual G2P model supporting Hindi, Gujarati, and Marathi
2. A phoneme clustering method that reduces output vocabulary
3. Quantitative comparison: baseline vs. clustered model
4. Visualizations of phoneme clusters (UMAP/t-SNE)
5. Reproducible pipeline and documentation